Response matrices for the FCC-ee correction workflow

FCC-ee Xsuite tuning

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Abstract

This note describes the response matrices implemented in this repository: closed orbit, normal optics and horizontal dispersion, coupling and vertical dispersion, tunes, and first- and second-order chromaticities. Equations use the same signs, coordinates, and knob units as the code. Analytical responses are local, uncoupled-optics approximations; numerical family responses are obtained from Twiss probes. The appendices explain the four-point Gauss–Legendre integration and the propagation of optics inside a quadrupole.

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1 Reference machine and matrix convention

`prepare_reference` in `correction_workflow.py` prepares a common reference with errors off, correction trims zero, and nominal nonlinear strengths (`on_nonlinear=1`). It computes periodic 4D Twiss at `delta0=0` with chromatic quantities enabled. Correction is performed without radiation or tapering. All response builders use this clean reference; the resulting matrices remain fixed during correction, while residuals are evaluated on the current machine.

Let $\boldsymbol{u}$ be the physical correction knobs, F_a an observable, and T_a its target. The stored matrix is

$$r_a(\boldsymbol{u}) = F_a(\boldsymbol{u}) - T_a, \qquad R_{aj} = \left.\frac{\partial r_a}{\partial u_j}\right|_{\text{ref}}, \qquad \boldsymbol{r}(\boldsymbol{u} + \Delta\boldsymbol{u}) \simeq \boldsymbol{r}(\boldsymbol{u}) + R\Delta\boldsymbol{u}. \tag{1}$$

Targets are independent of the correction knobs. Orbit targets are live BPM offset expressions: they change with the error ramp, but not with an orbit trim at a fixed ramp setting. Thus the stored responses are derivatives of the observables themselves.

Rows and columns follow the optimizer's target and vary lists. The matrix contains neither target weights nor knob scales. If W contains the target weights and S the `Vary.weight` scales, the solver uses

$$A = WRS, \qquad \Delta\boldsymbol{u} = -SA^{+}W\boldsymbol{r}. \tag{2}$$

Here A^{+} is the SVD pseudoinverse with the configured singular-value cutoff; only active rows and columns enter the solve. Step scaling and limits may shorten the resulting change. The workflow evaluates the changed machine and checks its correction goals separately.

Module	Observables	Response construction
Orbit (x, y)	x, y at BPMs	Closed-orbit Green function
Optics	$\beta_{x,y}, \alpha_{x,y}, D_x, Q_{x,y}$	Thick-quadrupole integrals
Coupling	$\Re f, \Im f, D_y$	Skew-coil midpoint model
Tune	Q_x, Q_y	Forward Twiss probes
Chromaticity	Q'_x, Q'_y	Central Twiss probes
Second order	Q''_x, Q''_y	Central Twiss probes

Throughout, $p \in \{x, y\}$ denotes a transverse plane, $\sigma_x = +1$, $\sigma_y = -1$, and Q_p is its tune. The phase $\mu_p(s)$ is measured in *turns*, as in Twiss, so its radian value is $2\pi\mu_p$. All optics functions in the following kernels are reference quantities. The longitudinal coordinate is s; δ is the relative momentum deviation.

2 Closed-orbit response

For observation s_i and a positive kick θ_{pj} at s_j, the periodic uncoupled Green function is

$$G_p(s_i, s_j) = \frac{\sqrt{\beta_p(s_i)\beta_p(s_j)}}{2\sin(\pi Q_p)} \cos[2\pi|\mu_p(s_i) - \mu_p(s_j)| - \pi Q_p]\,. \tag{3}$$

`orbit_response` returns $R_{ij} = G_p(s_i, s_j)$, with rows at BPMs and columns at correctors. Knobs represent kicks in radians. Their installation uses Δ`knl[0]` $= -\theta_x$ and Δ`ksl[0]` $= +\theta_y$, so a positive knob produces a positive canonical transverse momentum kick in either plane.

The matching residual is $x_i - d_{x,i}$ or $y_i - d_{y,i}$, where $d_{p,i}$ is the live BPM offset. The stored matrix is in the laboratory frame and has no cross-plane blocks. The orbit acceptance diagnostic additionally rotates the offset-subtracted coordinates by the BPM roll; that diagnostic rotation is not included in the analytical matrix.

3 Normal-quadrupole response

3.1 Distributed strength and kernels

For quadrupole j of length L_j, let $u_j = \Delta k_{1j}$ be a uniform normal-gradient trim in m^{-2}. A small slice $\mathrm{d}\ell$ produces the focusing perturbation $\sigma_p u_j \mathrm{d}\ell$. Write its location as $t_j(\ell) = s_j + \ell$ and define

$$d_p(i,t) = \mu_p(s_i) - \mu_p(t), \qquad \Phi_p(i,t) = 4\pi|d_p(i,t)| - 2\pi Q_p. \tag{4}$$

The implementation in `normal_quadrupole_response` integrates the following kernels along each magnet:

$$\frac{\partial \beta_p(s_i)}{\partial k_{1j}} = -\frac{\sigma_p \beta_p(s_i)}{2\sin(2\pi Q_p)} \int_0^{L_j} \beta_p(t_j(\ell)) \cos \Phi_p(i, t_j(\ell))\, \mathrm{d}\ell, \tag{5}$$

$$\frac{\partial \alpha_p(s_i)}{\partial k_{1j}} = -\frac{\sigma_p}{2\sin(2\pi Q_p)} \int_0^{L_j} \beta_p(t_j(\ell)) \Big[\operatorname{sgn} d_p(i, t_j(\ell)) \sin \Phi_p(i, t_j(\ell)) + \alpha_p(s_i) \cos \Phi_p(i, t_j(\ell)) \Big], \mathrm{d}\ell, \tag{6}$$

$$\frac{\partial Q_p}{\partial k_{1j}} = \frac{\sigma_p}{4\pi} \int_0^{L_j} \beta_p(t_j(\ell))\, \mathrm{d}\ell. \tag{7}$$

In particular, the beta and alpha rows are derivatives of the *ordinary* Twiss functions, not of logarithms or normalized-alpha combinations. The cosine term proportional to $\alpha_p(s_i)$ in Eq. (6) is needed for the ordinary-alpha response. Relative-beta weighting is applied later: a target weight proportional to $1/\beta_{p,\mathrm{ref}}(s_i)$ makes its weighted residual proportional to $\beta_p/\beta_{p,\mathrm{ref}} - 1$.

A normal trim gives a horizontal momentum kick $\Delta p_x = -u_j x \mathrm{d}\ell$. On the clean reference orbit, the corresponding dispersion source is $-u_j D_x(t)\mathrm{d}\ell$, giving

$$\frac{\partial D_x(s_i)}{\partial k_{1j}} = -\int_0^{L_j} G_x(s_i, t_j(\ell)) D_x(t_j(\ell))\, \mathrm{d}\ell. \tag{8}$$

These are linearized, uncoupled, on-momentum response formulas. They do not include a coupled-optics or nonlinear feed-down response of the dirty machine. The integrals are evaluated using the same four internal sample points for beta, alpha, tune, and dispersion (Appendices A and B).

3.2 From magnet derivatives to actual knobs

The low-level matrix $B_{aj} = \partial F_a / \partial k_{1j}$ has columns for individual thick quadrupoles. Its rows are grouped by quantity: β_x, β_y, α_x, α_y, D_x, Q_x, Q_y. Beta, alpha, and dispersion can have different observation lists.

`build_normal_quad_optics_response_matrix` in `corr_optics.py` finds the quadrupoles controlled by each independent knob through xdeps. Let K_{1j} be the magnet's total integrated normal strength, obtained from `get_total_knl_ksl()[0][1]`. For knob u_b and its configured probe step h_b, the strength slope is measured without running Twiss:

$$C_{jb} \simeq \frac{K_{1j}(u_b + h_b) - K_{1j}(u_b - h_b)}{2h_b}, \qquad R_{ab} = \sum_j \frac{B_{aj}}{L_j} C_{jb}. \tag{9}$$

Each knob is restored after the probes, also on failure. The factor $1/L_j$ converts a distributed-gradient response into an integrated-strength response. This handles knobs shared by several quadrupoles, derived family expressions, and contributions to the total strength beyond the main `k1` field. Integrated changes are modeled as distributed uniformly over each magnet. The supported magnets are unsliced thick quadrupoles with fixed positive lengths. The matrix is reordered into the actual target list before use.

The standard optics optimizer has five rows per observation, $(\beta_x, \beta_y, \alpha_x, \alpha_y, D_x)$, followed by Q_x, Q_y. The low-level helper also returns

$$s_j^{\mathrm{quad}} = \left[\max\left(\int_0^{L_j} \beta_x(t_j(\ell))\mathrm{d}\ell, \int_0^{L_j} \beta_y(t_j(\ell))\mathrm{d}\ell \right) \right]^{-1}. \tag{10}$$

These auxiliary scales are currently discarded by `corr_optics.py`; its standard optics varies use unit weights.

4 Skew-quadrupole response: coupling and vertical dispersion

4.1 Physical groups and midpoint optics

Each coupling knob u_g is an integrated skew-quadrupole strength in m^{-1}, installed on one physical sextupole. If the magnet has several pieces with lengths L_m, the added skew strength is

$$\Delta\texttt{ksl[1]}_m = u_g \frac{L_m}{L_g}, \qquad L_g = \sum_{m \in g} L_m. \tag{11}$$

The analytical model replaces the distributed coil by one kick at the physical magnet midpoint $s_g = s_{g,0} + L_g/2$. It uses the entrance of the first piece and the total length, assuming the pieces describe one contiguous magnet. For $\ell_g = L_g/2$, midpoint optics are approximated by drift transport:

$$\beta_{p,g} = \beta_{p,0} - 2\alpha_{p,0}\ell_g + \frac{1+\alpha_{p,0}^2}{\beta_{p,0}}\ell_g^2, \tag{12}$$

$$\mu_{p,g} = \mu_{p,0} + \frac{1}{2\pi}\,\mathrm{atan2}(\ell_g, \beta_{p,0} - \alpha_{p,0}\ell_g), \tag{13}$$

$$D_{x,g} = D_{x,0} + \ell_g D_{p_x,0}. \tag{14}$$

Here D_{p_x} denotes `dpx`. This is a clean-orbit, straight-sextupole approximation; unlike the normal-quadrupole response, it uses no Gauss points.

4.2 Resonance-driving terms

For an observation s_i, define the forward phase from source to observation with the ring wrap chosen by longitudinal ordering:

$$\Delta\mu_p(i,g) = \mu_p(s_i) - \mu_{p,g} + Q_p\,\mathbf{1}_{s_g > s_i}. \tag{15}$$

The complex coupling derivatives implemented in `skew_quadrupole_response` are

$$\frac{\partial f_{1001}(s_i)}{\partial u_g} = \frac{\sqrt{\beta_{x,g}\beta_{y,g}}\,e^{2\pi\mathrm{i}[\Delta\mu_x(i,g) - \Delta\mu_y(i,g)]}}{4[1 - e^{2\pi\mathrm{i}(Q_x - Q_y)}]}, \tag{16}$$

$$\frac{\partial f_{1010}(s_i)}{\partial u_g} = \frac{\sqrt{\beta_{x,g}\beta_{y,g}}\,e^{2\pi\mathrm{i}[\Delta\mu_x(i,g) + \Delta\mu_y(i,g)]}}{4[1 - e^{2\pi\mathrm{i}(Q_x + Q_y)}]}. \tag{17}$$

Their real and imaginary parts form four real response blocks. These signs and phases follow the code's positive `ksl[1]` and Twiss RDT conventions.

A positive integrated skew trim produces $\Delta p_y = u_g x$, so its vertical dispersion response is

$$\frac{\partial D_y(s_i)}{\partial u_g} = \frac{\sqrt{\beta_y(s_i)\beta_{y,g}}}{2\sin(\pi Q_y)}\cos[2\pi\Delta\mu_y(i,g) - \pi Q_y]\,D_{x,g}. \tag{18}$$

The helper initially groups rows by quantity; `corr_coupling.py` reorders them into five rows per observation: $(\Re f_{1001}, \Im f_{1001}, \Re f_{1010}, \Im f_{1010}, D_y)$. Targets are the corresponding reference values. Column scales are $s_g^{\mathrm{skew}} = 1/\sqrt{\beta_{x,g}\beta_{y,g}}$ and are supplied through `Vary.weight`, rather than multiplied into the stored matrix.

During construction the code checks one distributed probe, $u_g = 10^{-5} s_g^{\mathrm{skew}} \sin(g + 0.3)$ with zero-based group index g. It compares the Twiss change with $R\boldsymbol{u}$, both for the full vector and for its D_y block, and rejects relative norm discrepancies above 5%. The clean zero trims are restored afterwards. This directional check tests the combined response, not every column independently.

5 Numerical family responses

These matrices are computed by perturbing family trims and evaluating Twiss on the clean machine. All dependent magnets follow their existing expressions. Probes use the reference closed-orbit particle as `co_guess` and the shared transfer-matrix and chromatic-step settings. Knobs are restored after each probe sequence, including when an evaluation fails.

5.1 Tunes

The two tune trims are additive offsets to `kqf2` and `kqd1`, in m^{-2}. Unlike the analytical tune rows in the optics matrix, `build_tune_response` measures their full family response with a forward probe of $h = 10^{-7}\,\mathrm{m}^{-2}$:

$$R^{Q}_{pb} = \frac{Q_p(\boldsymbol{u}_0 + h\boldsymbol{e}_b) - Q_{p,\mathrm{ref}}}{h}. \tag{19}$$

This is a 2×2 matrix, evaluated with `chrom=False`.

5.2 First-order chromaticity

Define $Q'_p = \mathrm{d}Q_p/\mathrm{d}\delta$, returned as `dqx` and `dqy`. Four additive sextupole-family offsets control `ksf1`, `ksf2`, `ksd1`, and `ksd2`, in m^{-3}. `build_chromaticity_response` returns the 2×4 matrix

$$R^{Q'}_{pb} = \frac{Q'_p(\boldsymbol{u}_0 + h\boldsymbol{e}_b) - Q'_p(\boldsymbol{u}_0 - h\boldsymbol{e}_b)}{2h}, \qquad h = 10^{-4}\,\mathrm{m}^{-3} \quad \text{by default.} \tag{20}$$

Each probe uses `chrom=True`; the targets are the shared reference chromaticities at nominal nonlinear strengths.

5.3 Second-order chromaticity

The two knobs `dko_arc_f` and `dko_arc_d` represent k_3 offsets in m^{-4}. They act on the quadrupoles controlled by the two arc quadrupole families. For each affected element the installer adds Δ`knl[3]` $= u_b L_j$ (with its correction switch), preserving the existing multipole expression. The 2×2 response is

$$R^{Q''}_{pb} = \frac{Q''_p(\boldsymbol{u}_0 + h\boldsymbol{e}_b) - Q''_p(\boldsymbol{u}_0 - h\boldsymbol{e}_b)}{2h}, \qquad h = 0.01\,\mathrm{m}^{-4} \quad \text{by default.} \tag{21}$$

Twiss returns Q''_p as `ddqx` and `ddqy`. The reported quadratic coefficient is instead $Q''_p/2$, since

$$Q_p(\delta) = Q_p(0) + Q'_p\delta + \tfrac{1}{2}Q''_p\delta^2 + \cdots. \tag{22}$$

The optimizer stores derivatives and targets of Q'' without dividing by two. Consequently a tolerance of one on $Q''/2$ corresponds to a target tolerance of two on `ddqx` or `ddqy`.

For both chromatic responses, the Twiss momentum step is `delta_chrom=2e-4`. This momentum step is distinct from the family-knob probe h. It is shared by reference and probe calculations; the earlier 10^{-6} momentum step gave unreliable second derivatives in this workflow. Both chromatic corrections are used in refinement, after the error ramp, with first-order correction preceding second-order correction.

6 Implementation map and scope

The analytical kernels are in `response_builders.py`. Assembly and target ordering are in `corr_orbit.py`, `corr_optics.py`, and `corr_coupling.py`. The numerical derivatives are in `corr_tune.py`, `corr_chromaticity.py`, and `corr_second_order_chrom.py`.

The unweighted matrices are installed with `set_response` by `CorrectionOptimizer`. Evaluations and the correction sequence belong to `FCCeeCorrection`.

The analytical formulas describe a stable periodic reference away from their resonant denominators. They are not exact responses of a strongly coupled, misaligned, nonlinear, radiating machine. Finite-difference Twiss tests in `tests/test_fccee_optics_response.py` check normal-optics responses, including compound knobs and distinct beta/alpha locations. The coupling directional check provides an additional runtime consistency test. Four-point quadrature improves the longitudinal integration of the chosen model; it does not remove that model's approximations. Neither response validation nor correction convergence establishes dynamic or momentum acceptance.

A Four-point Gauss–Legendre integration

For each quadrupole integral, the code calls `numpy.polynomial.legendre.leggauss(4)`. The returned nodes ξ_n are the roots of the fourth Legendre polynomial on $[-1, 1]$, and w_n are its quadrature weights:

Node ξ_n	Weight w_n
−0.861136311594053	0.347854845137454
−0.339981043584856	0.652145154862546
+0.339981043584856	0.652145154862546
+0.861136311594053	0.347854845137454

For a magnet of length L_j, map these dimensionless nodes and weights to physical distances and integration lengths:

$$\ell_{jn} = \frac{L_j}{2}(1 + \xi_n), \qquad W_{jn} = \frac{L_j}{2} w_n, \qquad \int_0^{L_j} f(\ell)\,\mathrm{d}\ell \simeq \sum_{n=1}^{4} W_{jn} f(\ell_{jn}). \tag{23}$$

Thus $\sum_n W_{jn} = L_j$, and W_{jn} has units of metres. The rule is exact for polynomials through degree seven. Our kernels contain transported optics and trigonometric phase factors, so their integration is still approximate. No adaptive subdivision or quadrature error estimate is currently performed.

For example, Eq. (5) becomes

$$B_{\beta_p(i),j} \simeq -\frac{\sigma_p \beta_p(s_i)}{2\sin(2\pi Q_p)} \sum_{n=1}^{4} W_{jn} \beta_p(t_j(\ell_{jn})) \cos \Phi_p(i, t_j(\ell_{jn})). \tag{24}$$

The same nodes enter the other normal-quadrupole kernels. One evaluates the *whole response integrand* at each point; averaging beta and phase first would not give the same result. In particular, beta and phase vary inside a strong quadrupole, so one point or an entrance/exit average can miss that variation. Gauss integration retains four samples while avoiding extra Twiss calculations: local optics are obtained by the transport below.

B Optics propagation inside a quadrupole

B.1 Constant-gradient transfer matrix

Let $P = 1 + \delta_j$ at the quadrupole entrance and $K_p = \sigma_p k_{1j}/P$. Within the model, k_{1j} and P are constant. Define

$$C(\ell) = \begin{cases} \cos(\sqrt{K_p}\ell), & K_p > 0, \\ 1, & K_p = 0, \\ \cosh(\sqrt{-K_p}\ell), & K_p < 0, \end{cases} \tag{25}$$

$$S(\ell) = \begin{cases} \sin(\sqrt{K_p}\ell)/\sqrt{K_p}, & K_p > 0, \\ \ell, & K_p = 0, \\ \sinh(\sqrt{-K_p}\ell)/\sqrt{-K_p}, & K_p < 0. \end{cases} \tag{26}$$

In canonical transverse coordinates (q, p_q), the matrix is

$$M_p(\ell) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} C & S/P \\ -PK_pS & C \end{pmatrix}. \tag{27}$$

Its determinant is one. In the zero-gradient limit, $a = d = 1$, $b = \ell/P$, $c = 0$, giving a drift. In the nominal on-momentum reference $P = 1$; the helper nevertheless retains the entrance momentum factor in its local transport. This transport alone does not make the response kernels general off-momentum formulas.

B.2 Beta, alpha, phase, and dispersion

With entrance values β_0, α_0 and $\gamma_0 = (1 + \alpha_0^2)/\beta_0$, covariance transport gives

$$\beta(\ell) = a^2\beta_0 - 2ab\alpha_0 + b^2\gamma_0, \tag{28}$$

$$\alpha(\ell) = -ac\beta_0 + (ad + bc)\alpha_0 - bd\gamma_0, \tag{29}$$

$$\gamma(\ell) = c^2\beta_0 - 2cd\alpha_0 + d^2\gamma_0. \tag{30}$$

The helper only needs $\beta(\ell)$; local alpha is not an input to the response kernels. The $\alpha(s_i)$ in Eq. (6) is the value at the observation point, read directly from the reference table.

The phase increment follows from

$$\sin\Delta\psi = \frac{b}{\sqrt{\beta_0\beta(\ell)}}, \qquad \cos\Delta\psi = \frac{a\beta_0 - b\alpha_0}{\sqrt{\beta_0\beta(\ell)}}. \tag{31}$$

Accordingly, the code uses

$$\mu(\ell) = \mu_0 + \frac{\operatorname{atan2}(b, a\beta_0 - b\alpha_0)}{2\pi}. \tag{32}$$

The two-argument arctangent retains the quadrant. The implementation uses its principal branch without further unwrapping, assuming the modest phase advance of an individual lattice quadrupole; it is not a general transport routine for arbitrarily large internal phase advances.

For a straight quadrupole in this clean, uncoupled model, the dispersion is transported homogeneously:

$$\begin{pmatrix} D(\ell) \\ D_p(\ell) \end{pmatrix} = M_p(\ell) \begin{pmatrix} D_0 \\ D_{p,0} \end{pmatrix}, \qquad D(\ell) = aD_0 + bD_{p,0}. \tag{33}$$

D_p is the derivative of canonical momentum with respect to δ (`dpx` or `dpy`), not an independently assumed slope coordinate. There is no bending source inside this transport model. Only $D(\ell)$ is needed at the integration samples.

The code vectorizes these calculations over all quadrupoles and four samples, returning arrays of shape $(N_{\text{quad}}, 4)$ for beta, phase, integration weight, and dispersion. It uses the main `k1` field for this internal transport. The separate knob-to-strength mapping in Eq. (9) accounts for changes in total integrated normal strength.